UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

PART A (10 MARKS)

Answer ALL questions.

1. Which operator sends data to cout?
   A. <<
   B. =
   C. ==
   D. >>

2. Which keyword prevents a variable's value from being changed after initialisation?
   A. break
   B. static
   C. void
   D. const

3. Which phase of the program development life cycle defines what the program must do?
   A. Analysis
   B. Coding
   C. Testing
   D. Maintenance

4. Which part of a for loop is executed after each iteration?
   A. The condition only
   B. The loop body twice
   C. The initialisation
   D. The update expression

5. What does a compiler do?
   A. Translates the whole source program before execution
   B. Draws flowcharts automatically
   C. Executes one source statement at a time
   D. Stores only integer values

6. Which structure chooses between two alternative paths?
   A. array
   B. struct
   C. if...else
   D. for

7. Which data type stores a whole number?
   A. float
   B. string
   C. char
   D. int

8. Which loop is guaranteed to execute its body at least once?
   A. while
   B. if
   C. do...while
   D. for

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9. What are parallel arrays?
   A. Arrays whose elements at the same index are related
   B. Arrays with no index
   C. Arrays that hold only characters
   D. Arrays stored in different programs

10. Which values can a bool variable hold?
   A. any integer
   B. any decimal
   C. any character
   D. true or false

PART B (60 MARKS)

QUESTION 1

This question tests the program development life cycle.

a) Define source code and object code.

(3 marks)

b) Describe THREE stages of the program development life cycle.

(2 marks)

QUESTION 2

A program reads a positive integer N and calculates the sum and average of the numbers 1 to N.

a) Draw a flowchart for the sum and average calculation.

(8 marks)

b) Write pseudocode and a complete C++ program for the algorithm.

(5 marks)

QUESTION 3

This question tests declarations, constants and expressions.

a) Write C++ declarations for: an integer age, a double price, a constant discount rate of 0.10 and a character grade.

(6 marks)

b) Write C++ statements to read age and price, calculate price after discount, and display the result.

(5 marks)

QUESTION 4

Study the following array program segment:

```
int a[5] = {2, 4, 6, 8, 10};
int s = 0;
for (int i = 0; i < 5; i++) {
   s += a[i];
}
cout << s;
```

a) Trace the program and state the output.

(7 marks)

b) Rewrite the segment using a do...while loop that sums only values greater than 5.

(8 marks)

QUESTION 5

A class records ten student marks in an array.

a) Declare an integer array of ten marks and read the marks from the user.

(8 marks)

b) Write a program segment to calculate and display the average, highest mark and number of marks above the average.

(8 marks)

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PART C (30 MARKS)

QUESTION 1

A catering company charges RM12.50 per meal. Orders of 50-99 meals cost RM11.50 each, 100-199 cost RM10.50 each, and 200 or more cost RM9.50 each.

Write a complete C++ program that reads the customer name and quantity, validates the quantity, calculates the total, displays a receipt, and repeats until the user chooses to stop. At the end, display the total quantity and total sales.

(15 marks)

QUESTION 2

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

END OF QUESTION PAPER